Aidan Marley

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EDUCATION

University of Rochester **Rochester, NY**
B.S. in Mechanical Engineering, Minor in Aerospace Engineering *Expected Graduation May 2027*

- 3.5 GPA, Dean's List (Fall 2023, Fall 2025, Spring 2026)
- Varsity Soccer (Aug 2023–Dec 2024): Balanced 30+ hrs/week with coursework; UAA All-Academic Honoree (2023–24, 2024–25)
- *Highlighted Coursework:* CFD, FEM, Vibrations, Fluid Mechanics, Heat Transfer, Thermodynamics, Heat Power Application, Mechanical Design, Circuits & Signals, Instrumentation & Measurement, Fabrication

RELEVANT WORK EXPERIENCE

Mechanical Engineering R&D Intern, *Energetiq Technology (subsidiary of Hamamatsu Photonics)* *Summer 2026,* **Wilmington, MA**

- Owned design of water-cooled next-gen lamp head for an LDLS® system; built and validated a SolidWorks Flow Simulation CFD heat-transfer model, setting boundary conditions, refining meshes and running parametric sweeps of ambient, chiller and thermal-load conditions.
- Achieved superior thermal management over fan-cooled baseline; design expected to improve laser stability while meeting design for manufacturability (DFM) goals; passed project management review.
- Owned separate design of fan-cooled lamp head attachment to regulate surface temp, minimize acoustic output, and address particle and stability concerns; compared fan CFM ratings and custom intake air funnel geometries using SolidWorks Flow Simulation.
- Created SolidWorks models and detail and exploded-view assembly drawings for engineering change orders (ECOs); managed revisions in PDM.

Engineering Intern, *Cianbro Corporation* *Summer 2025,* **Portsmouth Naval Shipyard, Kittery, ME**

- Supported structures team on $2.2B NAVFAC Mid-Atlantic P-381 Multi-Mission Dry Dock #1 marine infrastructure project.
- Performed sideloading / buoyancy analyses to determine concrete pour requirements and calculate water shim slip conditions.
- Created schematics and technical plans for inspections, pumps and generator use to ensure proper management of cofferdam water levels.
- Created and presented daily operating plans (PODs) to executives and crews, coordinating positions of 20+ vessels and cranes around tides, currents, project needs and safety.

Boatshop Assistant, *Woodman Boats (boatbuilder)* *Part-time, 2019–2026,* **Kennebunkport, ME**

- Gained hands-on exposure to traditional wooden hull construction, structural timber framing, mechanical repairs, and machining.

SELECT ENGINEERING PROJECTS

Sailboat Hull CFD: Displacement-to-Planing Transition *Fall Semester 2026,* **Rochester, NY**

- Reconstructing 19-ft Lightning-class hull from lines plans; digitizing offsets in MATLAB and modeling parts in NX.
- Setting up Ansys Fluent free-surface (VOF) CFD simulations across a speed sweep to map drag, trim and sinkage, and pinpoint where the hull transitions from displacement to planing.
- Results expected December 2026; invited to publish findings in the International Lightning Class Association newsletter.

Pneumatic Engine Design & Fabrication *Fall Semester 2026,* **Rochester, NY**

- Designing, building and bench-testing a pneumatic engine as a solo class project; modeling the assembly in Siemens NX.
- Analyzing kinematics and loads in NX Motion Simulation, and structural and vibration response in NX Nastran FEM.
- Designed casting pattern and cast flywheel; generated NX toolpaths and G-code to CNC-machine flywheel and backbone; manually turned and milled piston and crankshaft; injection-molded feet, formed sheet-metal base and laid up composite propeller.

Bio-Inspired Wind Turbine Blade Design & Testing: Tubercle Aerodynamics *Spring Semester 2026,* **Rochester, NY**

- Designed and tested whale-inspired leading-edge tubercles, increasing power output 30% at low Reynolds numbers near cut-in speed.
- Owned blade design: generated NACA 63-215 tubercle geometry in VBA and MATLAB, matching control blade volume within 0.15%.
- Modeled control and tubercle blade sets in SolidWorks, then sliced and 3D-printed them; built turbine assembly.
- Ran NX FEA and CFD to evaluate structural and aerodynamic performance, correlating results with test data.
- Co-developed test protocol and conducted wind tunnel testing that verified the power increase.
- Awarded Best Fluids Design Project, Mechanics Laboratory (ME 241).

Translation Stage Prototype *Fall Semester 2025,* **Rochester, NY**

- Designed and implemented closed-loop 2D motion control system by integrating LVDTs, servomotors, cam disks, Arduino microcontroller, and MATLAB to accurately position a translation stage.
- Developed and calibrated sensing and actuation models, including voltage-displacement fits, and encoded them into custom MATLAB control functions with range and linearity safeguards.
- Wrote MATLAB program that converted an image into scaled millimeter coordinates and automatically drove the stage to reproduce image in dotted and continuous-line forms.

TECHNICAL SKILLS

CFD & Thermal/Fluids: Ansys Fluent, SolidWorks Flow Simulation, Siemens NX CFD (Simcenter 3D Flow/Thermal)
FEA & Structural: Siemens NX Nastran (static, modal, thermo-structural, topology optimization), Ansys Mechanical, Abaqus
CAD & Drafting: SolidWorks, Siemens NX (incl. Motion Simulation), Creo, GD&T (ASME Y14.5), SolidWorks PDM
Programming & Software: MATLAB, Python, C++, VBA, Microsoft Office (Excel, Word, PowerPoint, Outlook), SharePoint
Testing: Wind Tunnel, Simcenter Testlab, Strain Gauges, Thermocouples, Arduino, Uncertainty Analysis
Fabrication: 3D Printing (FDM, PolyJet), CNC Programming & Machining, Sheet Metal, Welding, Composite Layup, Injection Molding, Casting